

Model Paper - Database Management Systems

Generated: 2026-02-26 00:09:44

Q1 (20 marks)

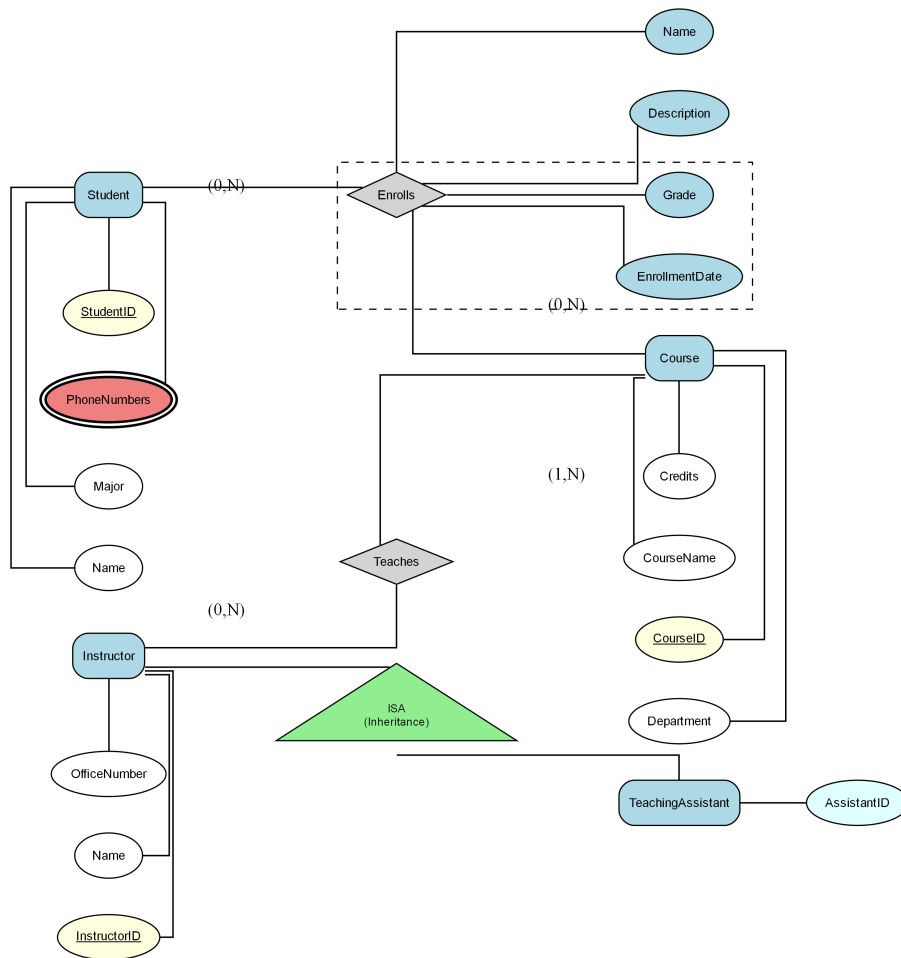
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

In a university database, there are four main entities: Student, Course, Instructor, and Enrollment. The Student entity includes attributes: StudentID, Name, Major, and PhoneNumbers. The Course entity has attributes: CourseID, CourseName, Credits, and Department. The Instructor entity consists of InstructorID, Name, and OfficeNumber. connects Students and Courses with attributes EnrollmentDate and Grade, indicating when a student enrolled in a specific course and their resulting grade. Each Student can enroll in multiple Courses, and each Course can have many Students. Instructors teach Courses, and a Course is taught by only one Instructor. Additionally, Instructors may have courses as a specialization through an ISA relationship with a TeachingAssistant subclass, which has the attribute AssistantID. A Student can be related to many Courses, and a Course can be related to many Students through the 'Enrolls' relationship (Student participation is optional, Course participation is optional). A Instructor can be related to many Courses through the 'Teaches' relationship (Instructor participation is optional)

Figure: EER



- Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)
- What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies F over R :
 $F = \{ \{AB \rightarrow C, C \rightarrow D, D \rightarrow E, E \rightarrow A, AC \rightarrow B\} \}$.

- Find all the keys in relation R using the attribute closure method. (4 marks)
- Is R in 3NF? Give reasons for your conclusion. (3 marks)
- Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A library is developing a comprehensive database system to manage its book inventory and member details. Various roles have been designated for the maintenance and management of this system. Laura is the chief database administrator responsible for the overall design and upkeep of the database. Steve and Rachel serve as junior database administrators reporting to Laura, where Steve oversees the books database and Rachel manages the members database. Eric works as a database

developer under Steve and Rachel, focusing on creating and managing the schemas for books and members. David and Sophia are data entry operators tasked with entering information about books and members into the system.

a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

b) Code Segment: `java String sql = "SELECT * FROM employees WHERE department = ?";
PreparedStatement pstmt = connection.prepareStatement(sql); pstmt.setString(1, "IT"); ResultSet rs =
pstmt.executeQuery();` Which type of JDBC statement is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)

c) Briefly explain the options available for transferring data between two database tables and between different servers. (2 marks)

d) Briefly explain how authentication and authorization are achieved in Oracle Database. (3 marks)

e) A library is developing a comprehensive database system to manage its Thi inventory and Thi details. Different roles are assigned to various team This for database administration and management. Laura is the chief database administrator responsible for the overall management of the database system's design and upkeep. Steve and Rachel are junior database administrators under Laura's supervision. Steve manages the This database, while Rachel handles the This database. Eric is a database developer reporting to Steve and Rachel. His responsibilities include designing and implementing database schemas for This and This. David and Sophia are data entry operators responsible for inputting Thi information and Thi records into the system. Sarah is the senior database administrator tasked with overseeing the entire database system's creation and maintenance.. Nathan is a database developer responsible for designing and implementing database schemas.. Michael is a data entry operator responsible for inputting data into the system.. Emily is a junior database administrator managing the inventory database..

- i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)
- ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)
- iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the Loans database using a user-defined server role. (5 marks)
- iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)
- v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the Members and Loans databases. (3 marks)

Q4 (40 marks)

Consider the following schema of a database designed for a Hospital: Patient (patientId: int, name: varchar(100), phoneNumber: varchar(15), dob: date) Doctor (doctorId: int, name: varchar(100), specialty: varchar(50), phoneNumber: varchar(15)) Appointment (appointmentId: int, patientId: int, doctorId: int, appointmentDate: date, status: varchar(20)) Fee (feeId: int, appointmentId: int, amount: real, paymentStatus: varchar(20)) The 'Patient' table stores information about patients with a unique ID, name, and date of birth. The 'Doctor' table holds details about doctors, including their specialties. The

'Appointment' table tracks appointments between patients and doctors. The 'Fee' table records the fee amount and payment status for each appointment. The 'Appointment' table manages appointment records, tracking scheduled appointments between patients and doctors with dates and times. The 'Fee' table contains relevant information for the database system.

a) Write SQL Queries to perform the following: i. Find the name and phone number of the patient who has an upcoming appointment on the date '2023-12-01'.

i. Find the name and phone number of the patient who has an upcoming appointment on the date '2023-12-01'.
(4 marks)

ii. Find the patient who has the highest total amount compared to all other patients. Find the patientId and name.
(6 marks)

iii. Find the name and name from Doctor, Patient, and Appointment where Appointment.appointmentDate is not null. (7 marks)

b) b) Create a function to calculate the total fee amount for a given patient. This function should account for fee with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) c) Create a trigger that automatically updates the 'amount' column in the 'Fee' table whenever a new fee record is added, or an existing fee record is updated. The 'amount' column should contain the total fee amount for each patient. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by patient. The patient pays only 50% of the total fee amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)